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Nicholas Wilkinson

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Factors influencing the small-scale distribution of House Sparrows *Passer domesticus* in a suburban environment

NICHOLAS WILKINSON

The Maltings, 99B North End, Meldreth, Cambridgeshire, SG8 6NX, UK

Capsule Suburban House Sparrows prefer gardens with native bushes but show no detectable response to garden insecticide use.

Aims To identify the factors that influence House Sparrow distribution among suburban gardens.

Methods Bicycle transects were used to survey House Sparrows in front gardens in suburban Oxford in the spring of 2004. A subset of 498 gardens was selected comprising all those where sparrows were found and an equal number of randomly selected sparrow-free gardens. In each of these gardens the density of bushes and the proportion of native bushes were recorded and insecticide use and house age were sampled by questionnaire. Binary logistic regression was used to compare these variables with the presence or absence of House Sparrows. A larger scale analysis was also conducted comparing 25 squares, each 250 m². The density of House Sparrows in the total area of front gardens in these squares was compared to the area of green space within the square, its distance from the city edge and encounter rates of corvids on the cycle transects.

Results House Sparrows show a preference for gardens with native bushes. They also prefer gardens with a higher overall density of bushes but appear to make no distinction based on insecticide use or house age. There is no evidence that green space, distance from the city edge or density of corvids affect sparrow numbers at a larger spatial scale.

Conclusion House Sparrows might benefit from planting of native bushes but not apparently from a reduction in garden insecticide use. The preference for native vegetation most likely reflects the associated abundance of insect food, which is required by the growing chicks. Further investigation of the habits of House Sparrows in the breeding season may shed light on the causes of the recent decline of the species.

In recent decades populations of the House Sparrow *Passer domesticus* have declined severely across western Europe (Summers-Smith 1993). In Britain the species has declined by over 50% since 1976 (Siriwardena *et al.* 2002), warranting its placement on the national Red Data list (Gregory *et al.* 2002).

Numerous hypotheses have been put forward to explain the decline, which has engendered deep public concern. These include population explosions of cats, corvids or Sparrowhawks *Accipiter nisus*, scarcity of nest-sites in modern buildings, increased pollution or a decline in food supplies (see Crick *et al.* 2002b for a review). One popular hypothesis (Summers-Smith 1999, Nicholson 2000) is that there is insufficient insect-food to feed young chicks. In support of this Mitschke *et al.* (1999) showed that House Sparrow

breeding success in central Hamburg is apparently limited by aphid numbers. However, a demographic analysis based on Nest Record Scheme and ring-recovery data (Freeman & Crick 2002) suggests that the decline in Britain was caused by a drop in the survival rates of first-year and, subsequently, adult birds rather than chicks (Crick *et al.* 2002b).

Bland *et al.* (2004) have shown that suburban gardens in Britain support higher densities of breeding birds than was previously believed. For House Sparrows suburban areas are the most important habitat, supporting almost 35% of the British population at much higher densities than are found in urban or rural areas (Siriwardena *et al.* 2002). Although rural gardens support even higher densities, they cover a much smaller total area. These results support earlier findings by Heij & Moeliker (1986) and Earlé (1986) in individual cities in The Netherlands and South Africa.

Email: nicholas.m.wilkinson@gmail.com

However, suburban House Sparrow populations have also witnessed the most severe declines (Siriwardena *et al.* 2002), constituting a greater proportion of the overall national decline than those in any other habitat (Crick *et al.* 2002b). Although particularly severe declines have occurred in certain city centres (Sanderson 1996, Dott & Brown 2000, Prowse 2002, Summers-Smith 2000), urban and rural populations as a whole appear to have declined less severely.

Suburban House Sparrow populations also show the lowest breeding performance (Crick & Siriwardena 2002) and a recent rise, which is believed to have slowed the rate of decline in other habitats, has not occurred in suburbia (Freeman & Crick 2002). This contrasts with the work of Heij & Moeliker (1986) and Earlé (1986) who found breeding performance to be highest in the suburbs, and one of Heij & Moeliker's two rural study sites appeared to be a 'sink' (*sensu* Pulliam 1988) supported by immigrant birds from suburbia. However, demographic analysis shows that a drop in breeding performance is unlikely to be a primary cause of the national decline (Crick *et al.* 2002b).

The work of Crick *et al.* (2002a) is invaluable in outlining the national patterns of the House Sparrow's decline. However, a national analysis is probably insufficient to identify a single cause, given the fine-grained population dynamics of the species (Hole 2001) and the strong variation in the timing and magnitude of the decline among different habitats and regions of Britain (Siriwardena *et al.* 2002). Small-scale studies, particularly in suburban areas, are also needed (Crick *et al.* 2002b).

That few such studies have so far been conducted perhaps reflects the difficulty of surveying House Sparrows, particularly in suburban environments (Bland *et al.* 2004, Bland 1998; Churcher & Lawton 1987, Crick *et al.* 2002b). The only study to analyse sparrow distribution at the scale of individual houses and gardens (Tully 2000, 2001) found that roofing tile type affected nest-site selection. Bland (1998) and Paston (2000) both found that housing estates held the highest densities of House Sparrows, which might also reflect differences in nest-site availability or levels of urban hygiene (Siriwardena *et al.* 2002). Bland (1998) also found apparent negative correlations between the densities of sparrows and Feral Pigeons *Columba livia*, Magpies *Pica pica* and Carrion Crows *Corvus corone*, but this result has not yet been subjected to statistical analysis.

The aim of this paper is to quantify the breeding

season habitat associations of suburban House Sparrows in Oxford, UK, at the fine scale of individual gardens and within the broader suburban environment. Particular attention is paid to their distribution with respect to those variables suggested as factors in the national decline. A secondary aim is the development of a technique capable of providing valid survey data for this skulking and gregarious species within the highly heterogeneous and largely inaccessible environment of suburbia.

METHODS

Study area

The city of Oxford (1°12'W, 51°46'N) covers approximately 30 km², most of which can be classified as suburban development. Suburban spread began in the late eighteenth century but much suburban housing dates from the 1920s and 1930s (Day 1976).

Rural studies of House Sparrow populations (Hole *et al.* 2002, Easterbrook 1999) have been conducted nearby. In southeast Britain as a whole House Sparrows are found at higher densities than in other regions of the country but have also declined more severely (Siriwardena *et al.* 2002).

Twenty-five 250-m squares were selected at random for sampling from the Ordnance Survey National Grid within suburban Oxford. 'Suburban' squares were defined as those comprising over 50% residential housing and lying outside the city centre.

Survey method

Because much appropriate habitat is inaccessible by virtue of lying within private back gardens (Bland *et al.* 2004), only front gardens were surveyed for sparrows (but see Paston 2000). The problem of the clumped distribution and skulking behaviour of the birds proved more intractable. Preliminary observations suggested that obtaining accurate estimates of numbers when more than five sparrows were congregated in dense vegetation would be impractical. The survey method was therefore designed to yield a comparative index of sparrow density.

Cycle transects (Bland 1998) were conducted along the entire length of streets and recognized cycle lanes within each square and sparrows were detected by sight or sound. At each garden where they were detected, three counts were made, at 30-second intervals, of all

birds immediately visible. Sparrows on rooftops above front gardens were included but those merely passing overhead were not.

Each square was surveyed three times between late April and late June 2000, within the breeding season of the species in Britain (Summers-Smith 1963). The survey timetable followed a blocked design with two squares being surveyed on most mornings, less than 2 hours and 40 minutes after sunrise. Surveys were postponed in the event of heavy rain. On two of the surveys for each square, counts were made of all members of the crow family (Corvidae), which are likely egg-predators of House Sparrows (Summers-Smith 1963).

As it was not possible to conduct detailed surveys of all 2319 front gardens covered by the cycle transects, a subset of the gardens within each square was selected. This comprised all gardens within that square where sparrows had been detected along with an equal number of randomly selected 'sparrow-free' gardens. In total 498 gardens were surveyed, half of which had been found to contain sparrows.

In November 2000 counts were made of bushes and trees within each of the survey gardens. A 'bush' was defined as any plant over 50 cm high with a structure sufficiently complex that less than 50% of its immediate background could be seen through its foliage. Climbing plants were included only if they had a significant three-dimensional structure of their own, rather than adhering closely to a wall. Bushes native to Britain were also counted (see Appendix).

A questionnaire was delivered to owners of all study gardens to survey insecticide use among residents and to ascertain the age of houses, which may be correlated with the number of potential nest-sites. The questionnaire also asked whether residents had seen sparrows in their gardens in the last 12 months.

Analysis

At all gardens where House Sparrows had been detected a mean was calculated of all nine counts, including zero counts. A comparative index of sparrow density for a square was calculated by summing these mean counts for all such gardens within a square and dividing this figure by the total number of gardens in the square.

Corvid density was calculated as the mean count of corvids for each square divided by the total cycle-transect length therein. The total 'green' area within each square and the distance of each square from the edge of the Oxford built-up area were both calculated using 1:1250 Ordnance Survey maps. The latter usually marked an interface with arable land. These three variables were compared to the index of sparrow density by multiple linear regression.

For each of the 498 survey gardens the following were calculated: the area of the garden (in m²), the density of bushes (number/m²) and the proportion of those bushes which were native.

For all gardens for which data were available, insecticide use was codified according to two separate systems (Table 1). Where data were available house age was given to the nearest decade.

These small-scale data were compared to the presence or absence of sparrows in front gardens using binary logistic regression. Gardens were blocked according to their distribution among the 25 large-scale study sites. The logarithmic transformation was applied to garden area and the angular transformation to the proportion of native bushes.

Due to the incomplete returns a smaller data set was available for questionnaire than for field data and separate analyses were therefore conducted (see Table 2).

Table 1. Questionnaire survey on insecticide use in front gardens.

Insecticide use*	No. households	Proportion of all respondents (%)
System 1		
Insecticide use	54	23.1
No insecticide use	171	73.1
System 2		
Would never use insecticide	117	50
Do not use insecticide but would do under certain circumstances	57	24.4
Use insecticide in the period April–May more than once a year	19	8.1
Use insecticide in the period April–May less than once a year	29	12.4
Use insecticide but never in the period April–May	0	0

*Insecticide use is characterized according to two systems, the second being designed to distinguish householders who deliberately withhold insecticide from those whose gardens naturally lack insect pests, and to determine the level of use and whether it occurs in the House Sparrow breeding season.

Table 2. The five separate binary logistic regression analyses run on the small-scale data. Separate analyses were necessary due to missing data and the inevitable correlation between bush density and proportion of native bushes.

Model formula	<i>n</i>	Designed to test
$P = X + A + B + N + I_1 + H$	140	Effect of house age
$P = X + A + B + N + I_1$	225	Effect of insecticide (use versus non-use)
$P = X + A + B + N + I_2$	225	Effect of insecticide (five-category system)
$P = X + A + B + N$	498	Effects of all other variables
$P = X + A + B + N$	458	Effects of all other variables within the subset of gardens containing bushes

Variables: P, presence/absence of sparrows; X, block (study square); A, garden area; B, bushes/m; N, proportion of native bushes; I, insecticide use (categorical); H, house age.

Table 3. Level of response to the questionnaire and its various sections. All owners of the 498 study gardens received a questionnaire but not all were returned. On most returned questionnaires some questions were left blank.

	Number of residents responding	Percentage of all residents	Percentage of respondents
Questionnaire return	234	47	100
Insecticide use	225	45	96
House age	179	36	76
Presence or absence of sparrows in last 12 months	192	38	82

Large-scale data were analysed using MINITAB Version 13.3 and small-scale data in SPSS version 11.5.

RESULTS

Questionnaire return

Levels of response to the questionnaire are summarized in Table 3. Answers to the question about sparrows seen in the last 12 months tend to agree with the survey results (binomial test, $n = 200$, $P = 0.006$, based on a z approximation). However, 56 respondents (29%) reported seeing sparrows when none were detected on the survey and 18 (9%) reported the absence of sparrows where the survey had detected them.

Large-scale data

Results of the analysis are summarized in Table 4. There is a positive relationship between the density of House Sparrows and corvids within a square (Fig. 1) although the wide confidence limits of the coefficient call its strength into question. House Sparrow density is apparently unrelated to 'green' area or distance from the city edge. Removal of a single outlier produced a significant, though weak, negative relationship between green area and House Sparrow density. There are no firm grounds for its removal.

Small-scale data

Gardens with a higher density of bushes were more likely to contain sparrows (Wald = 8.66, $n = 498$, $P = 0.003$, Fig. 2) as were gardens with a greater proportion of native bushes (Wald = 10.74, $n = 498$, $P = 0.001$, Fig. 3). There was no significant effect of garden area (Wald = 1.29, $n = 498$, ns).

Because bush density and proportion of native bushes are correlated, in that where the former is zero the latter is also zero, the above result could be spurious. However, a further analysis conducted on the set of gardens containing at least one bush (see Table 2) produced a similar result.

Insecticide use did not appear to influence the presence or absence of sparrows either when considered

Table 4. Variables affecting the density of sparrows within front gardens in the twenty-five 250-m study squares; summary of multiple linear regression analysis. For fuller explanation of variables see text.

	<i>F</i> ratio	<i>P</i> value	Coefficient*	<i>r</i> ² (%)
'Green' area	2.64	0.119	$-0.07 \times 10^{-6} \pm 0.08 \times 10^{-6}$	9.57
Distance from edge	1.59	0.221	$0.02 \times 10^{-4} \pm 0.03 \times 10^{-5}$	5.85
Corvid density	5.32	0.031	$0.38 \times 10^{-4} \pm 0.32 \times 10^{-4}$	19.3

*Coefficients given with 95% confidence intervals. The very low values merely reflect the low values taken by the index of sparrow density (see text).

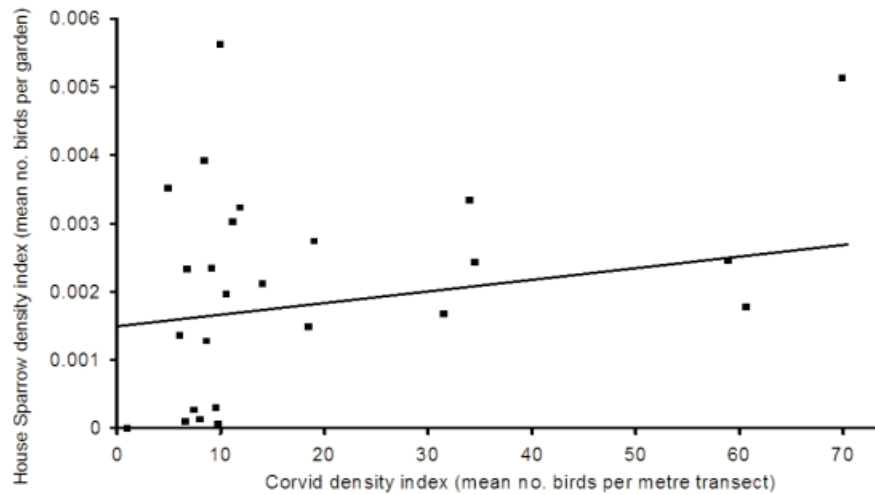


Figure 1. Relationship between densities of House Sparrows and corvids within the 25 study squares in suburban Oxford. Densities are calculated from survey data; see text for details.

as use versus non-use (Wald = 0.38, $n = 225$, ns) or under the five-category scheme in Table 1 (Wald = 1.57, $n = 225$, ns). House age also did not appear to influence sparrows (Wald = 0.256, $n = 140$, ns).

DISCUSSION

It is hoped that the methods employed in this study will be of use to future researchers. It is encouraging that a strong correlation was found between the survey method and the questionnaire data on sparrow sightings. Questionnaire return rates of close to 50% are also encouraging considering that, in contrast to the British Trust for Ornithology/CJ Wildbird Foods Ltd Garden Birdwatch and many other public participation surveys (Royal Society for the Protection of

Birds 2003), respondents were not selected on the basis of known interest in ornithological or wildlife issues. Judging by the comments made on many questionnaires the high level of response from local residents reflects widespread concern about the House Sparrow's decline.

Large-scale data

Large-scale data should be treated with some caution because, as the survey only covered front gardens, the total area surveyed varied among the 25 squares. This variation might bias survey results (Gaston *et al.* 1999) but the possibility is relatively slight.

House Sparrows appear to be more abundant in areas with more corvids, a result that conflicts with the

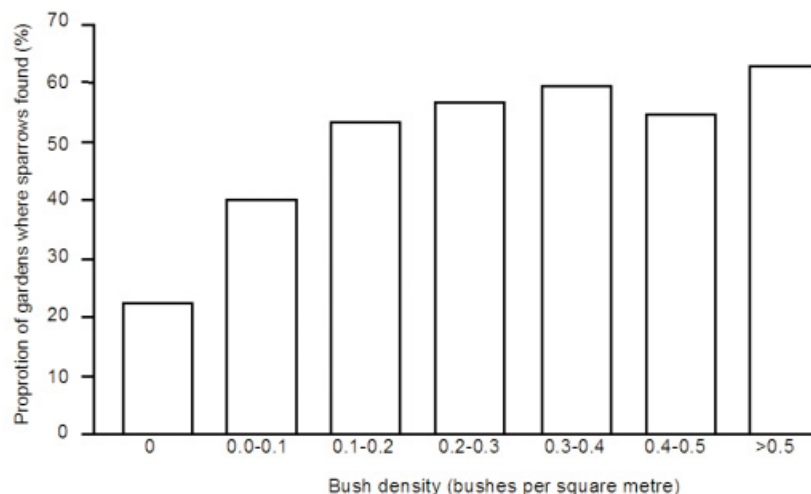


Figure 2. Comparative frequency of House Sparrow occurrence in front gardens with varying densities of bushes.

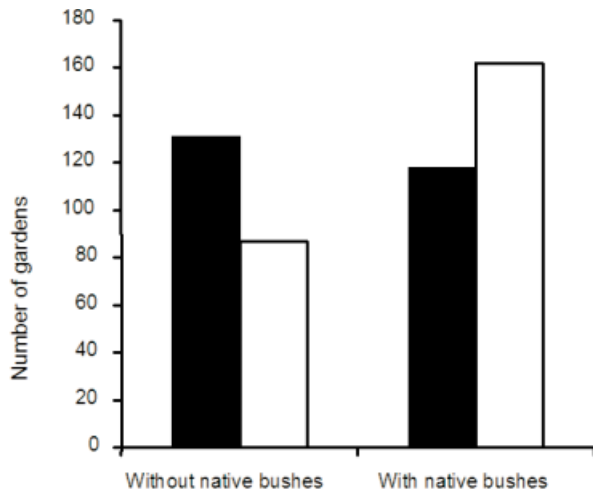


Figure 3. Comparative frequency of House Sparrow occurrence among sampled gardens with and without native bushes. This graph is intended to show the effect of native bushes and is not a true representation of the proportion of gardens containing sparrows, which is in fact much lower than suggested here. ■, Without sparrows; □, with sparrows.

findings of Bland (1998) in Bristol. This pattern may be an artefact of the survey method, arising from the fact that more survey time was spent in House Sparrow-rich squares. However, such extra survey time was spent counting sparrows in front gardens at close range and, as most corvids were seen either in large open spaces or while flying overhead, it is unlikely that increased survey time alone gave rise to the significantly greater counts. This result probably indicates a genuine correlation but is almost certainly influenced by other confounding variables and provides no evidence either to support or refute the hypothesis that egg predation by corvids is responsible for the decline in sparrow populations. Notably neither green area nor distance from the city edge can be the confounding variable responsible for this pattern as both have been eliminated statistically.

Neither of these two variables seems to affect House Sparrow abundance within suburbia. This contrasts with the finding of Heij & Moeliker (1986) in Rotterdam that a positive correlation existed between abundance and green area. However, the difference made by the elimination of a single datum casts doubt on any conclusions that might be drawn and highlights the problems of small sample size ($n = 25$). It is also possible that distance to the nearest city edge is not always a meaningful measure given that the Oxford built-up area is a highly irregular shape; continuous corridors of undeveloped land connect city-centre parks with arable fields.

Small-scale data

It is not surprising that House Sparrows show a strong preference for gardens with dense vegetation, given their well-known use of bushes for forage and shelter (Summers-Smith 1963). The robustness of this result is increased by the fact that it is achieved in spite of the reduced detectability of birds in dense vegetation.

It also appears that House Sparrows prefer native vegetation. Siriwardena *et al.* (2002) mention a hypothesis that a reduction in the planting of native bushes is the cause of the House Sparrow's decline but this study does not support this hypothesis. Out of 498 gardens surveyed, 118 had native bushes but no sparrows, and native bushes are therefore unlikely to be a limiting resource. However, sparrows may have been present in these 118 gardens but not detected on the survey, or the particular species of bushes there might not have provided suitable forage or shelter. The possibility that reduced planting of native bushes could have affected sparrow numbers in Oxford cannot therefore be ruled out.

Planting of native bushes for the benefit of garden birds has been suggested on the grounds that native vegetation supports more insects (Cannon 1999). Although House Sparrows are predominantly herbivorous, insects are vital in the diet of the growing chicks (Summers-Smith 1963) and this explanation is therefore plausible. An alternative explanation, that exotic plant material is less palatable to the sparrows themselves, is less likely. House Sparrows are extremely catholic in their choice of vegetable food (Summers-Smith 1963) and presumably subsist on local vegetation in all regions to which they have been introduced (i.e. on all continents except Antarctica). This study therefore suggests that the distribution patterns of adult sparrows during the breeding season in suburban Oxford are affected by insect abundance. In order to verify these conclusions, further studies might investigate whether the preference for native bushes remains outside the breeding season.

Further investigation of small-scale habitat preferences within the breeding season might also help to determine the cause of the House Sparrow's decline. It appears that sufficient insects cannot be obtained on searches optimized for vegetable foods. This fact may affect the survival not only of chicks, but also of adults. Adult mortality is highest during the breeding season (Heij & Moeliker 1990, Crick & Siriwardena 2002, Summers-Smith 1988) and may be influenced by increases in energy expenditure or exposure to preda-

tion resulting from changes in small-scale habitat use.

Garden use of insecticide, on the other hand, is unlikely to have been a major factor in the decline of the House Sparrow, as it does not seem to affect choice of forage. The analysis using the category system in Table 1 allowed distinction between different types of insecticide use but still gave no significant result. Intrinsic problems of the questionnaire approach are less easily solved: a continuous variable is represented by categorical data and there is the possibility of bias in responses, perhaps resulting from an effort to appear more environmentally responsible. Also, if intensive use occurs mainly in back gardens it would not be detected by this survey. It is unlikely, however, that such biases would obscure any major effects. Certainly it appears that the density of native vegetation has a greater effect on House Sparrows than does garden insecticide use, at least at the levels found in suburban Oxford.

Little can be said regarding the importance of nest-site availability. The surrogate variable of house age gave no significant result but many questionnaire respondents commented that re-roofing or loft conversion had probably greatly reduced the number of available nest-sites in their homes. Tile type (Tully 2000, 2001) and roofing insulation (R.L. Bland, unpubl. data) may also be important. The age of a house is therefore probably not a valid indicator of its suitability as a House Sparrow nest-site and further work is needed to devise an appropriate index.

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(MS received 3 July 2003; revised MS accepted 10 December 2004)

APPENDIX

Bush (sic) species accepted as native on the surveys.

Common name	Scientific name
Common Yew*	<i>Taxus baccata</i>
Traveller's Joy	<i>Clematis vitalba</i>
London Plane*†	<i>Platanus ?hispanica</i>
Wych Elm	<i>Ulmus glabra</i>
English Elm*†	<i>U. procera</i>
Common Beech	<i>Fagus sylvatica</i>
English Oak	<i>Quercus robur</i>
Sessile Oak	<i>Q. petraea</i>
Silver Birch	<i>Betula pendula</i>
Downy Birch	<i>B. pubescens</i>
Common Hornbeam*	<i>Carpinus betulus</i>
Common Hazel	<i>Corylus avellana</i>
Limes*‡	All <i>Tilia</i> species were included
Black Poplar and hybrids*‡	<i>Populus nigra</i>
Blackcurrant	<i>Ribes nigrum</i>
Redcurrant†	<i>Ribes rubrum</i>
Gooseberry	<i>Ribes uva-crispa</i>
Brambles	<i>Rubus fruticosus</i> agg., and allied species (including <i>R. aescius</i>)
Roses†	All <i>Rosa</i> species, which are mostly of hybrid stock
Wild Cherry*	<i>Prunus avium</i>
Bird Cherry*	<i>Prunus padus</i>
Blackthorn	<i>Prunus spinosa</i>
Plums, damson & bullace†	<i>Prunus ?domestica</i>
Pear†	<i>Pyrus communis</i>
Apples††	All forms of <i>Malus</i> were included excepting recently introduced ornamental ‘crab-apples’
Rowan	<i>Sorbus aucuparia</i>
Whitebeams	<i>Sorbus aria</i> agg., and allied species (including <i>S. rupicola</i>)
Medlar†	<i>Mespilus germanica</i>
Hawthorns	<i>Crataegus monogyna</i> and <i>C. laevigata</i>
Common Broom	<i>Cytisus scoparius</i>
Holly*	<i>Ilex aquifolium</i>
Box	<i>Buxus sempervirens</i>
Sycamore†	<i>Acer pseudoplatanus</i>
Field Maple	<i>A. campestre</i>
Ivy	<i>Hedera helix</i>
Ash	<i>Fraxinus excelsior</i>
Elder	<i>Sambucus nigra</i>
Guelder Rose	<i>Viburnum opulus</i>
Honeysuckle	<i>Lonicera periclymenum</i>

*It was not possible to definitely separate these native forms from similar exotic congeners; however, in all cases the native form is much the most commonly planted in gardens. †Long-established introductions. ‡Species complexes included. Privet *Ligustrum* and *Cotoneaster* species were not included as, in both cases, the native species is much rarer in gardens than exotic forms.